



3.6 Vertigo

3.6.1 Balance Assessment

◆ Key Features

- Electronystagmography (ENG)
- Rotational chair
- Dynamic post urography
- Vestibular evoked myogenic potential (VEMP)

Balance involves three components: (1) sensory input from the visual, vestibular, and proprioceptive systems; (2) central nervous system integration of these sensory signals with the subsequent generation of appropriate motor commands; and (3) adequate musculoskeletal abilities to perform the motor tasks. Proper assessment of balance/vertigo issues involves looking at all three of these components.

◆ Epidemiology

Dizziness is a common symptom in ~30% of people over the age of 65. Approximately 615,000 persons in the United States have been diagnosed with Ménière's disease. Between 10 and 64 persons per 100,000 are affected by benign paroxysmal positional vertigo (BPPV) each year. Hundreds of thousands of hospital days are incurred every year in the United States due to vertiginous symptoms.

◆ Clinical

Persons with vestibular disorders report a variety of symptoms. They can include, but are not limited to, vertigo (i.e., spinning sensation), dysequilibrium (i.e., subjective sense of falling), imbalance (i.e., observable unsteadiness), lightheadedness (i.e., feeling of faintness), oscillopsia (i.e., vision instability with head movement), nausea, and vomiting. Dizziness can also be accompanied by hearing loss, tinnitus, and hyperacusis (sound sensitivity). Dizziness can be brief (lasting seconds to minutes) or prolonged (lasting hours to days). Patients may report episodic or positional dizziness. Precipitating factors such as recent upper respiratory tract infection, head trauma, vestibulotoxic drug exposure, and migraine can be ascertained.

◆ Evaluation

Physical Exam

A full head and neck with otologic and cranial nerve exam is performed. Neurologic exam can also give insight into central causes of dizziness. Use of Frenzel goggles can help accentuate nystagmus on exam by removing visual fixation. Physical exam should include evaluation of spontaneous, gaze, and headshake nystagmus. Gait is assessed for ataxia and wide stance. Romberg and Fukuda stepping tests are done to evaluate for sway. A Dix-Hallpike test will assess presence of posterior canalolithiasis. Cerebellar testing (such as finger-to-nose) can test for dysmetria and tremor. A head impulse test can demonstrate catch-up saccades. Proprioception of upper and lower extremities can be assessed with a tuning fork. Mobility of the neck (e.g., cervicogenic dizziness) and vertebral artery (e.g., vertebral artery stenosis) should be tested as well.

Basic Balance Testing

ENG/VNG

Electronystagmography (ENG) and videonystagmography (VNG) test peripheral and central vestibular function. During ENG, electrodes are placed above and below the eyes, at the outside corner of the eye, and on the forehead. Electrical potentials during testing reflect direction and velocity of eye movements. VNG does not rely on electrical potentials but rather on video analysis of eye motion. VNG measures the movements of the eyes directly through infrared cameras and may be more accurate, more consistent, and more comfortable for the patient than traditional ENG. One advantage to VNG is the ability to test torsional nystagmus in addition to vertical and horizontal nystagmus. There are four main components to ENG/VNG:

1. Oculomotor activity: Smooth pursuit tracking, saccade analysis, gaze fixation, spontaneous nystagmus, and optokinetic stimulation
2. Positioning tests: Such as Dix-Hallpike
3. Positional tests: Nystagmus on head supine, head left and right, body left and right, and 30° incline

4. Caloric irrigation: Warm and cold air or water*Rotary Chair*

Rotational chair testing expands the evaluation of the peripheral vestibular system beyond the frequency and intensity limitations of the ENG/VNG evaluation. The phase, gain, and symmetry of the vestibular ocular reflex are assessed. The patient is then subjected to various rotations through sinusoidal harmonic acceleration tests and step tests with the head restrained to a chair with a computer-controlled motor in a darkened enclosure. Off-axis rotation provides ear-specific information and tests the utricle and superior vestibular nerve function. Standard rotational chair tests the horizontal semicircular canal function and stimulates the otoliths. Utricular function can specifically be assessed through subjective visual vertical or horizontal testing. The patient attempts to set an illuminated light to true vertical or horizontal in the absence of a visual reference and ambient light.

Posturography

Dynamic posturography assesses the functional relative use of vision, vestibular, and somatosensory cues. Sensory organization is assessed by measuring a patient's ability to maintain balance while disrupting somatosensory and/or visual input. The patient is exposed to six conditions using a combination of normal, eyes closed, and the tilting of the support surface and/or the visual surround. Instability or fall on conditions 5 (eyes closed, moving platform, stable visual surround) and 6 (eyes open, moving platform and visual surround) suggest peripheral vestibulopathy.

VEMP

Cervical vestibular evoked myogenic potentials (cVEMP) are recorded by electrodes placed on the ipsilateral sternocleidomastoid muscle in response to click stimuli. This assesses saccular function and the integrity of the inferior division of the vestibular portion of the eighth cranial nerve. Lower thresholds and increased peak-to-peak amplitudes suggest superior semicircular canal dehiscence. Asymmetry ratios $> 40\%$ suggest a saccular or inferior vestibular nerve weakness.

Ocular vestibular evoked potentials (oVEMP) are recorded by electrodes placed on the contralateral inferior oblique muscle in response to sound stimulus. This test assesses utricular function and the integrity of the superior division of the vestibular nerve. Lower thresholds and increased peak-to-peak amplitudes suggest superior semicircular canal dehiscence. Asymmetry ratios $> 40\%$ suggest a utricular or superior vestibular nerve weakness.

Other Tests

An audiologic evaluation should be conducted when a balance disorder is suspected to be vestibular in origin.

Pathology

Balance disorders can be caused by disorders of the peripheral vestibular system or the central vestibular system, systemic issues with parts of the body other than the head and brain, and vascular disorders. Peripheral vestibular system issues can be caused by Ménière's disease, labyrinthitis, BPPV, perilymphatic fistula, vestibular neuritis (VN), ototoxicity, vestibular schwannomas, superior canal dehiscence syndrome (SCDS), autoimmune disease, and other disorders. Central vestibular system issues can be caused by migraine, hydrocephalus, brain tumors, multiple sclerosis, cerebrovascular disease including transient ischemic attack or stroke, stress, tension, fatigue, vision disturbances, and other disorders. Systemic issues can include peripheral neuropathies, hyperventilation, and dehydration. Vascular disorders can include orthostatic hypotension, arteriosclerosis, vertebral artery stenosis, and vasovagal syndrome. Cervicogenic dizziness can occur after neck injury or in association with neck pain.

◆ Treatment Options

Treatment for the underlying cause of dizziness should be initiated. Factors that contribute to dizziness (e.g., migraine, visual disturbance) should be addressed and treated. In cases of BPPV, canalith repositioning is highly effective (e.g., the Epley maneuver for posterior canalolithiasis). Ablation of the vestibular organ through vestibulotoxic injections (e.g., intratympanic gentamicin) or surgery (e.g., labyrinthectomy or vestibular nerve section) may be appropriate in certain situations to allow symptom reduction through compensation. In cases of superior semicircular canal dehiscence or a perilymphatic fistula, surgery to repair the defect may alleviate the patient's symptoms. Neurology consultation is ordered when central vestibulopathy or complex vestibular migraine is suspected.

Vestibular therapy is helpful in a number of vestibular disorders. Adaptation, habituation, substitution, and particle repositioning exercises are addressed during vestibular therapy. Vestibulosuppressants can be prescribed for severe vertigo; however, this may prolong time to recovery.

◆ Complications

Vision problems and limitations in physical mobility can influence the ability to conduct accurately some components of ENG/VNG, rotary chair, posturography, and VEMP testing. Results can also be influenced by a variety of medications and alcohol.

3.6.2 Benign Paroxysmal Positional Vertigo

◆ Key Features

- Transient, episodic vertigo is induced by head position changes.
- Benign paroxysmal positional vertigo (BPPV) is most commonly caused by otoliths displaced into the posterior semicircular canal.
- Posterior canalolithiasis produces a latent geotropic upbeating torsional nystagmus that fatigues with time on Dix-Hallpike test.
- Repositioning maneuvers depend on canal involved.

Benign paroxysmal positional vertigo (BPPV) is the most common form of vertigo reported by multidisciplinary balance centers. Displaced otoconia settle in the lowest part of the inner ear, the posterior semicircular canal. Head motion moves these crystalline masses, stimulating the neuroepithelium of the posterior semicircular canal. This causes patients to experience brief episodes of vertigo. Vertigo with latent and fatigable geotropic upbeating rotary nystagmus elicited by the Dix-Hallpike maneuver is diagnostic of posterior canalolithiasis, and most cases are treated successfully with the Epley repositioning maneuver. The horizontal canal is the second most commonly affected semicircular canal; anterior (or superior) canalolithiasis is rare.

◆ Epidemiology

BPPV is the most common form of dizziness. Typically, it occurs spontaneously in elderly patients; in younger patients, BPPV often presents after head trauma or viral labyrinthitis. Symptoms can resolve without treatment.

◆ Clinical

Signs

In posterior canalolithiasis, upbeating, geotropic rotatory nystagmus is noted when a patient is placed in the Dix-Hallpike position. There may be a short latency (delay) in the onset of nystagmus, and the duration is brief (less than 1 minute). Nystagmus reverses upon sitting upright (i.e., downbeating, ageotropic rotary nystagmus). The response fatigues with repetition. In horizontal canalolithiasis, geotropic or ageotropic nystagmus is demonstrated on supine roll test. In anterior canalolithiasis, downbeating torsional nystagmus toward the affected ear can be seen during Dix-Hallpike; it is controversial whether anterior canalolithiasis is a true entity.

Symptoms

The patient experiences a spinning sensation that lasts less than a minute, which is associated with head movement, most often when bending forward, looking upward, or lying down and rolling from one side to another. Patients

may also have mild, generalized imbalance. Symptoms may also include nausea and vomiting.

Differential Diagnosis

Most diagnoses can be eliminated on history alone. The differential diagnosis for dizziness lasting seconds includes perilymphatic fistula, BPPV, vertebrobasilar insufficiency, superior semicircular canal dehiscence, and cervicogenic dizziness. Dizziness lasting hours can suggest endolymphatic hydrops, Ménière's disease, and vestibular migraine. When symptoms last for days, labyrinthine concussion, labyrinthitis, and vestibular neuritis are suspected. Dizziness lasting months may indicate a retrocochlear mass, vestibulotoxicity, or cerebellar disorder. Central etiologies such as stroke and multiple sclerosis can also present with dizziness.

◆ Evaluation

Physical Exam

A full neurologic exam should be performed. A basic vestibular exam is described in **Chapter 3.6.1**.

Imaging

Patients with positioning nystagmus suggestive of canalolithiasis do not typically require imaging. Computed tomography (CT) of the temporal bones may be useful in evaluating other causes of dizziness, such as superior semicircular canal dehiscence or temporal bone trauma. Magnetic resonance imaging (MRI) of the brain with and without gadolinium may be useful in ruling out intracranial process or retrocochlear mass.

Labs

Laboratory evaluation is not useful in the diagnosis of BPPV.

Other Tests

Other causes of vertigo may be evaluated with electronystagmography or videonystagmography, vestibular evoked myogenic potentials, rotary chair, and posturography. Tullio phenomenon and Hennebert sign for fistula can also be performed. Audiologic testing is obtained in vestibular symptoms other than BPPV.

Pathology

Cadaveric temporal bone dissections and labyrinthine operations performed in patients with symptoms of BPPV have demonstrated the presence of small calcium crystals (thought to be displaced otoconia from the saccule or utricle) within the posterior semicircular canal. Changing head position causes movement of otoliths within the canal, inducing motion of the cupula. There has also been speculation that cupulolithiasis (presence of otolithic crystals trapped on the cupula) plays a role in some cases. The most common canal affected is the posterior semicircular canal. The second most common canal is the horizontal canal.

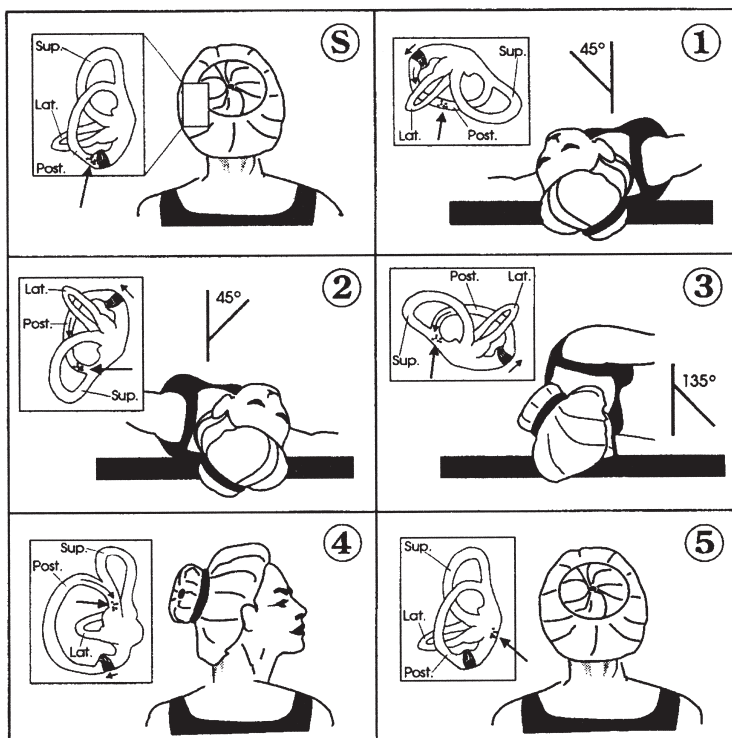


Fig. 3.12 The Epley repositioning maneuver. Positioning sequence for left posterior semicircular canal, as viewed by operator (behind patient). The inset boxes show exposed views of labyrinth, with migration of particles (large arrows). S, Start, patient seated. 1. Place head over end of table, 45° to the left. 2. Keeping head tilted downward, rotate 45° to the right. 3. Rotate head and body until facing downward 135° degrees from supine. 4. Keeping head turned right, bring patient to a sitting position. 5. Turn head forward, chin down 20°. Pause at each position until induced nystagmus approaches termination, or for T (latency + duration) seconds if no nystagmus. Keep repeating entire series (1 through 5) until there is no nystagmus in any position. (Used with permission from Epley JM. Particle repositioning for benign paroxysmal positional vertigo. *Otolaryngol Clin North Am* 1996;29:327.)

◆ Treatment Options

Medical

For posterior canal BPPV, the Epley repositioning maneuver is the mainstay of treatment, with success rates exceeding 90% (**Fig. 3.12**). This maneuver entails performing a Dix-Hallpike maneuver to elicit vertigo. Following cessation of vertigo, the patient's head is rotated 90° to the unaffected side. After 45 to 60 seconds, the head is then turned 90° further by having the patient lie on his or her side, such that the face is toward the floor. The

patient is then brought back to the upright position with his or her chin tucked to the chest. Each of these positions can be kept for approximately one minute. The canaliths are ultimately repositioned in the vestibule away from sensory epithelium. This procedure may be repeated as needed for relapses or refractory cases. Following the Epley, patients are instructed not to lie flat or assume any provoking position for a few days. Habituation exercises such as Brandt-Daroff exercises can be prescribed for relapsing cases or patients who cannot tolerate the Epley maneuver in the office. The Semont maneuver is an alternative repositioning maneuver for posterior canal BPPV.

Lateral canal BPPV can be treated with the barbecue roll (logroll), Gufoni, or Vannucci maneuvers. Anterior canal BPPV can be treated with an Epley maneuver on the opposite ear or modifications of the Semont maneuver. Habituation exercises can also be prescribed. Referral to vestibular therapy may be necessary when maneuvers cannot be performed in the office (e.g., cervical disease or reduced mobility).

Surgical

Two surgical procedures can be performed for refractory cases of posterior canal BPPV: singular neurectomy and posterior canal occlusion. Singular neurectomy entails sectioning the singular nerve (posterior ampullary nerve) as it courses from the ampulla of the posterior canal to join the sacular nerve to form the inferior part of the vestibular nerve. This procedure is technically difficult, carries a high risk of postoperative sensorineural hearing loss (SNHL), and is not often performed. Posterior canal occlusion involves identifying the posterior canal via a transmastoid approach and packing the canal with bone dust, fascia, or bone wax to create a barrier preventing the flow of endolymph and stimulation of neuroepithelium. Canal plugging can also be done in refractory cases of horizontal and anterior canal BPPV. When performed bilaterally, there is hypofunction of both canals that may cause oscillopsia.

◆ Outcome and Follow-Up

Following the Epley maneuver, patients are instructed not to lie flat or assume any provoking position for the next week, to allow canaliths to settle. Patients with persistent symptoms may have the Epley maneuver repeated or may benefit from Brandt-Daroff exercises or treatment with a trained vestibular physical therapist. BPPV after head trauma often affects multiple canals and is more difficult to treat; vestibular therapy referral is commonly obtained in these patients.

3.6.3 Ménière's disease

◆ Key Features

- Ménière's disease is classified into two categories: definite Ménière's disease and probable Ménière's disease (2015 Diagnostic Criteria).
- The classical presentation is episodic vertigo associated with low- to medium-frequency sensorineural hearing loss and fluctuating aural symptoms lasting 20 minutes to 12 hours.
- Ménière's disease is caused by excess endolymph within the labyrinth ("endolymphatic hydrops").
- Conservative, non-vestibular-destructive treatments are considered prior to invasive, vestibular-destructive procedures for controlling vertigo.

Ménière's disease, or endolymphatic hydrops, is a common cause of recurrent vertigo. The diagnostic criteria of Ménière's disease was revised in 2015. Ménière's disease is classified into two categories: definite Ménière's disease and probable Ménière's disease.

Definite Ménière's disease is defined as (1) two or more spontaneous episodes of vertigo each lasting 20 minutes to 12 hours, (2) audiometrically documented low- to medium-frequency sensorineural hearing loss in one ear, at least on one occasion, and (3) fluctuating aural symptoms in the affected ear (hearing, tinnitus, or fullness), with (4) no better vestibular diagnosis.

Probable Ménière's disease is defined as (1) two or more episodes of vertigo or dizziness each lasting 20 minutes to 24 hours and, (2) fluctuating aural symptoms in the affected ear (hearing, tinnitus, or fullness), with (3) no better vestibular diagnosis.

◆ Epidemiology

The prevalence of Ménière's disease is approximately 34 to 190 per 100,000 population. There is a slight female predominance, and age of onset ranges from the third to seventh decade. Bilateral Ménière's disease rarely presents simultaneously; one ear precedes the other by several years. Progression to bilateral Ménière's disease occurs in 5 to 33% of affected patients.

◆ Clinical

Signs and Symptoms

A classic presentation of Ménière's disease is episodic vertigo lasting minutes to hours, associated with fluctuating hearing loss, tinnitus, and aural fullness. Patients often report nausea and vomiting, which may be followed by severe fatigue. Between episodes, patients will commonly have resolution

Table 3.11 Diagnostic guidelines for Ménière’s disease

Definite Ménière’s disease (1) Two or more spontaneous episodes of vertigo each lasting 20 minutes to 12 hours (2) Audiometrically documented low- to medium-frequency sensorineural hearing loss in one ear, at least on one occasion (3) Fluctuating aural symptoms in the affected ear (hearing, tinnitus, or fullness) (4) No better vestibular diagnosis
Probable Ménière’s disease (1) Two or more episodes of vertigo or dizziness each lasting 20 minutes to 24 hours (2) Fluctuating aural symptoms in the affected ear (hearing, tinnitus, or fullness) (3) No better vestibular diagnosis

Data from American Academy of Otolaryngology—Head and Neck Foundation Inc. Guidelines for the diagnosis and evaluation of therapy in Meniere’s disease. Otolaryngol Head Neck Surg 1995;113(3): 181–185.

of symptoms. Occasionally, imbalance is reported. Familial Ménière’s disease should be considered if at least one first- or second-degree relative has definite or probable Ménière’s disease. Patients may report drop attack, also known as Tumarkin crisis, which presents as sudden spontaneous falls without loss of consciousness (patient remembers the episode).

Differential Diagnosis

The differential diagnosis for episodic vertigo is benign paroxysmal positional vertigo (BPPV), superior semicircular canal dehiscence syndrome (SCDS), perilymphatic fistula, autoimmune inner ear disease, vestibular schwannoma, endolymphatic sac neoplasm, other intracranial neoplasm, migraine-related vestibulopathy, and transient ischemic attacks (**Table 3.11**).

◆ **Evaluation**

Physical Exam

A complete neurologic examination should be performed. A physical examination may demonstrate the presence of low-frequency hearing loss via tuning fork exam. A basic vestibular exam with Tullio and Hennebert tests is performed to evaluate other causes of dizziness (described in **Chapter 3.6.1**). Nystagmus may be present during an acute attack; however, most patients with Ménière’s disease will have a normal exam at the time of an office visit.

Imaging

Patients with unilateral or asymmetric sensorineural hearing loss and dizziness should have a magnetic resonance imaging (MRI) study of the brain with and without contrast, including thin cuts of the internal auditory canals to rule out vestibular schwannoma or other retrocochlear pathology.

Labs

Autoimmune serology (e.g. antinuclear antibody [ANA], anti-neutrophil cytoplasmic antibodies [ANCA], rheumatoid factor [RF]), erythrocyte sedimentation rate (ESR), syphilis testing (rapid plasma reagin [RPR] or fluorescent treponemal antibody absorption [FTA-ABS]), diabetic testing, thyroid function tests, and Lyme titers (when appropriate) should be considered for patients exhibiting symptoms of bilateral Ménière's disease.

Other Tests

Audiograms are routinely used in the diagnosis and monitoring of Ménière's disease. Audiograms classically reveal hearing loss that is more severe in the low frequencies. Fluctuation may occur rapidly during acute episodes. Electrocochleography may aid in the diagnosis when the summing potential to action potential ratio is greater than 0.5. Electronystagmography or videonystagmography may demonstrate a significant vestibular weakness in the affected ear. Caloric testing is essential prior to consideration of ablative management.

Pathology

Ménière's disease is hypothesized to be caused by excess endolymph, which causes dilation of the stria vascularis and resultant ischemia. Ischemia then results in weakness in the walls of the scala media, causing rupture and mixing of endolymph with perilymph. Cadaveric temporal bone studies on patients with Ménière's disease demonstrate these findings.

◆ Treatment Options

Medical

Conservative and nondestructive approaches are initially tried in the treatment of Ménière's disease and include short-term vestibulosuppressants, instituting a low-salt diet (< 1500 mg of sodium daily), and potassium-sparing diuretics (Dyazide). Systemic steroids may be used in the short term to control acute attacks; intratympanic steroids have also been utilized for management of vertigo and associated sensorineural hearing loss (efficacy difficult to establish). Betahistine has also shown efficacy in managing vertigo episodes. Other nondestructive therapies include the Meniett device (Medtronic, Inc., Minneapolis, MN) for vertigo prevention.

Intratympanic gentamicin injection to medically ablate the labyrinth (40 mg/mL of buffered gentamicin solution) is a highly effective alternative to surgical ablation in patients with Ménière's disease who have debilitating vertigo. Gentamicin is toxic to dark cells as well as hair cells within the inner ear but preferentially affects vestibular cells. Hearing loss may occur in as many as 15% of patients. Multiple gentamicin injections may be required to ablate vestibular function completely in the affected ear; functional reserve can be monitored with caloric testing. Patients will report worse sensation of dizziness prior to improvement of symptoms. Vestibular training exercises may be helpful to improve balance following medical labyrinthectomy.

Surgical

Refractory Ménière disease may require more invasive treatments. The treatment algorithm depends on degree of residual hearing. Surgical procedures done in patients with serviceable hearing are endolymphatic sac decompression, with or without shunt placement, and vestibular nerve section with preservation of the cochlear nerve. When hearing is not serviceable, a labyrinthectomy or vestibulocochlear nerve section can be performed.

Endolymphatic sac surgery is performed via a transmastoid approach where the sac is identified and subsequently either decompressed or shunted into the mastoid cavity or arachnoid space. Vestibular nerve sectioning may be performed via a middle cranial fossa, retrosigmoid, or retrolabyrinthine approach. Labyrinthectomy may be performed via a transmastoid approach but results in hearing destruction. Vestibular nerve section and/ or labyrinthectomy (medical or surgical) should not be done bilaterally, as the patient will experience debilitating profound vestibular hypofunction and oscillopsia.

◆ Outcome and Follow-Up

The natural history of the disease appears to involve a gradual resolution of active symptoms in most patients over 5 years. Patients are generally followed clinically as needed. Serial audiometry to monitor stability of hearing is helpful, especially in patients with substantial fluctuation, and in patients who have been treated with aminoglycosides. Sudden hearing loss should be treated appropriately. Hearing loss is progressive over time.

3.6.4 Vestibular Neuritis

◆ Key Features

- In vestibular neuritis (VN), severe vertigo may last days to weeks.
- VN must be distinguished from acute intracranial pathology.
- Recovery may take several weeks.

Vestibular neuritis (VN) is a fairly common cause of acute-onset nonrecurring vertigo. The vertiginous episode is often preceded by viral upper respiratory tract infection. Inflammation of the vestibular nerve results in hypofunction of the affected side and causes vertigo. VN typically has no associated hearing loss.

◆ Epidemiology

Vestibular neuritis primarily affects the middle aged to the elderly but can occur at any age. No gender predominance has been noted.

◆ Clinical

Signs and Symptoms

Some cases of VN may be preceded by upper respiratory infection symptoms. Severe nausea and vomiting with dehydration may warrant hospital admission. Spontaneous and headshake nystagmus is usually present and can be more easily noted by using Frenzel goggles. Episodes usually last several days to weeks. Full compensation from a severe vestibular loss can sometimes take months. Interestingly, it is not unusual for some patients to develop benign paroxysmal positional vertigo (BPPV) several months after their VN symptoms resolve.

Differential Diagnosis

The differential diagnosis includes pathologies of both central and peripheral etiology. Cerebellar or brainstem infarction can cause sudden-onset vertigo. Peripheral causes of sudden-onset vertigo include perilymphatic fistula, Ménière's disease, BPPV, or lesions of the cerebellopontine angle. Bacterial or viral labyrinthitis can also mimic VN; however, in labyrinthitis patients generally report hearing loss as well and require additional treatment.

◆ Evaluation

Physical Exam

A complete neurotologic exam should be performed with Frenzel goggles. Spontaneous, gaze, and headshake nystagmus is generally present, is directed away from the affected ear and is suppressible. Due to a hypofunctioning labyrinth, patients will sway or fall to the affected side on Romberg testing or Fukuda stepping test. Patients will often present their heads tilted to one side, which often lessens their symptoms. A head thrust test will often be abnormal and indicates loss of the vestibuloocular reflex.

Imaging

Imaging is warranted to rule out intracranial infarction, hemorrhage, or other retrocochlear pathology. Magnetic resonance imaging (MRI) is the preferred study to identify infarction, but computed tomography (CT) can be used initially to rule out hemorrhage.

Labs

Lyme disease titers may be of value depending on geographic location.

Other Tests

An audiogram should also be performed to determine baseline hearing and assess for sensorineural hearing loss that may indicate another diagnosis. Electronystagmography or videonystagmography is not necessary but may be performed to confirm the unilateral vestibular weakness on the affected side, especially in patients with persistent dizziness and imbalance.

Pathology

No conclusive evidence of the exact etiology of VN exists. Cadaveric temporal bones have demonstrated inflammation and degeneration of primarily the superior vestibular nerve and the vestibular ganglion (Scarpa’s ganglion). Herpes virus has been demonstrated as a cause in some cases.

◆ Treatment Options

See **Table 3.12** for management options. As a rule, VN resolves spontaneously. The vertigo induced by the hypofunctioning vestibular nerve can resolve spontaneously through central compensation. Admission for severe vertigo and dehydration from persistent vomiting is sometimes required for rehydration and intravenous (IV) medication administration. Nausea may be symptomatically treated most effectively with vestibular suppressants: promethazine (25 mg parenterally–intramuscular or rectal suppository preferred; an IV route is discouraged because of severe reactions from inadvertent intraarterial infusion), meclizine (25 mg orally [PO]), or benzodiazepines such as diazepam (5 mg PO). Patients should be cautioned about using vestibular suppressants for long periods, as they can slow down central compensation.

Corticosteroids can be administered as methylprednisolone tablets (Medrol Dosepak, Pfizer, New York, NY) or a dexamethasone IV taper (Decadron 10 mg; Merck & Co., Inc., Whitehouse Station, NJ) and may improve long-term recovery. Acyclovir or valacyclovir (Valtrex; GlaxoSmithKline, Brentford, Middlesex, UK) may be utilized due to the association with

Table 3.12 Management options for vestibular neuritis

Vestibular suppressants
Promethazine (Phenergan, Pfizer, New York, NY) 25 mg IM, PR, PO every 6 hours as needed Diazepam (Valium; Roche Pharmaceuticals, Nutley, NJ) 2–10 mg IM, IV, PO two to three times daily as needed Clonazepam (Klonopin; Roche Pharmaceuticals, Nutley, NJ) 0.5–1 mg PO three times daily as needed
IV hydration
Lactated Ringer solution bolus or maintenance infusion
Steroids
Consider taper, e.g., methylprednisolone tablets (Medrol DosePak, Pfizer, New York, NY) 4-mg tabs PO, start 24 mg/day and taper by 4 mg daily over 6 days
Consider antivirals
Acyclovir 800 mg PO 5x daily x 1 week Valacyclovir (Valtrex, GlaxoSmithKline, Brentford, Middlesex, UK) 1,000 mg PO twice daily x 1 week
Vestibular physical therapy

Abbreviations: IM, intramuscularly; IV, intravenously; PR, per rectum; PO, orally.

herpes virus, but their efficacy is yet to be determined. Patients should be encouraged to perform at-home vestibular exercises. Vestibular rehabilitation can be helpful for patients with slow compensation.

◆ Outcomes and Follow-Up

Most patients never have recurrence of symptoms. Unsteadiness and dizziness may be experienced for months after an event. Some patients may go on to develop BPPV months later.

3.6.5 Migraine-Associated Vertigo

◆ Key Features

- Over a quarter of patients with migraine will report vestibular symptoms.
- The pathophysiology of vestibular migraine is poorly understood.
- Many patients do not experience headache in conjunction with attacks.
- Management focuses on treatment and prevention of migrainous attacks.

Migraine is among the most common central disorders that produce vestibular symptoms ranging from head motion intolerance to spontaneous vertigo. Although the association between migraine and vertigo has been well documented, a causal relationship has not been elucidated. The classification of this disorder includes vestibular migraine and probable vestibular migraines. Treatment is aimed at migraine control.

Vestibular migraine is defined as (1) at least 5 episodes with vestibular symptoms of moderate or severe intensity, lasting 5 minutes to 72 hours, (2) current or previous history of migraine with or without aura, and (3) one or more migraine features with at least 50% of the vestibular episodes (headache, photophobia/phonophobia, and visual aura), with (4) no better vestibular or International Classification of Headache Disorders (ICHD) diagnosis. Headaches should demonstrate at least two characteristics: one-sided location, pulsating quality, moderate or severe pain intensity, and aggravation by routine physical activity.

Probable vestibular migraine is defined as (1) at least 5 episodes with vestibular symptoms of moderate or severe intensity, lasting 5 minutes to 72 hours, and (2) history of migraine or migraine features during episodes, with (3) no better diagnosis.

◆ Epidemiology

Migraine is a common disorder that affects 18 to 29% of women and 6 to 14% of men. Vestibular migraine affects up to 1% of the general population. Familial occurrence of vestibular migraine with autosomal dominance has been documented in several families.

◆ Clinical

Signs and Symptoms

The typical migraine headache is unilateral, throbbing, moderate to severe, exacerbated by physical activity, and accompanied by sensitivity to light and noise. Migraines are classified with or without aura. Several vestibular symptoms can accompany migraine, including dizziness, head motion intolerance, and spontaneous vertigo. The attacks of vertigo can last minutes to days. Nausea and vomiting can also occur. Most patients are symptom-free between attacks, which can occur several times a month. Some patients will also experience other otoneurologic symptoms such as aural fullness. The attacks of vertigo are not temporally associated with headache symptoms during all episodes. In some women, there is an association with menstruation (i.e., perimenstrual migraine with vertigo). Benign paroxysmal vertigo of childhood is regarded as a precursor to migraine syndrome.

Differential Diagnosis

The differential diagnosis of episodic vertigo is broad and includes peripheral conductions such as benign paroxysmal positional vertigo, perilymphatic fistula, superior semicircular canal dehiscence, autoimmune inner ear disease, otosyphilis, and Ménière's disease. Central causes of recurrent vestibulopathy include basilar migraine, vertebrobasilar insufficiency, hydrocephalus, retrocochlear mass, and stroke. Anxiety, depression, and other underlying psychiatric disorder may cause dizziness as well; more than 50% of patients with vestibular migraine have comorbid psychiatric disorder.

◆ Evaluation

History

A thorough history focusing on vestibular and migrainous symptoms is critical. The definitions of vestibular migraine and probable vestibular migraines are as previously described. International Headache Society diagnostic criteria for migraine are described in **Table 3.13**. Migraine triggers

Table 3.13 International Headache Society (IHS) Diagnostic Criteria for Migraines

Recurrent headaches separated by symptom-free intervals and accompanied by any three of the following: • Abdominal pain • Complete relief after sleep • Nausea and/or vomiting • Aura • Hemicrania • Throbbing, pulsatile quality • Family history of migraine
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Data from International Headache Society. The International Classification of Headache Disorders. 2nd ed. *Cephalgia* 2004;24 (Supl 1).

are also important to identify. Other otologic symptoms, such as hearing loss, tinnitus, and aural fullness, may suggest a different or additional etiology for dizziness.

Physical Exam

The physical exam in patients with migraine-associated vertigo is usually normal. However, thorough head and neck examinations, including neurologic and vestibular exams, should be completed to exclude other causes of vertigo.

Other Tests

Vestibular testing can be helpful in separating peripheral from central causes of vertigo. Patients with migrainous vertigo usually have normal electronystagmography or videonystagmography tracings; however, abnormalities may be seen on rotary chair (e.g., increased vestibulo-ocular reflex time constant). Complete audiologic assessment should be obtained; low-frequency loss may be found as an early sign of Ménière's disease (a condition that can coexist with vestibular migraine).

Imaging

Magnetic resonance imaging (MRI) and computed tomography (CT) of the brain are usually normal in cases of migraine-associated vertigo but are useful in excluding other serious neurologic conditions. There is often considerable value in offering patients reassurance that MRI results are normal.

◆ Treatment Options

Treatment for migraine-associated vertigo centers on control of migrainous attacks by prophylactic or abortive medical management (**Table 3.14**). The choice of medications depends on the side effect profile and comorbidities of patients. Prophylactic treatment includes the avoidance of dietary triggers, β -blockers in patients with hypertension without asthma (propranolol 40 mg twice daily or metoprolol), calcium channel blockers (verapamil 80 mg three times daily or flunarizine), and anticonvulsants (topiramate, valproic acid, lamotrigine). If patients have anxiety, antidepressants (amitriptyline, nortriptyline, selective serotonin reuptake inhibitors) and anxiolytics (clonazepam) are considered. Acetazolamide is an effect in rare genetic disorders related to migraine-like episodic ataxia. Triptans (sumatriptan 25 mg PO at onset of symptoms) and analgesics (ibuprofen 600 mg) can be used to abort or reduce symptoms when they do occur.

Few studies have evaluated the effects of these management strategies on vestibular symptoms; however, there is some evidence that the same prophylactic drugs can be effective. For acute vertiginous episodes,

Table 3.14 Management strategies for migraine-associated vertigo

Prophylaxis	Treatments to abort acute attack
Avoidance of dietary triggers (e.g., aged cheeses, chocolate, beer, red wine, excessive caffeine, aspartame, monosodium glutamate, high salt)	Nonsteroidal anti-inflammatories: Ibuprofen 600 mg PO at onset
β-blockers: Propranolol 40 mg PO twice daily	For perimenstrual migraine with vertigo: Clonazepam (Klonopin; Roche Pharmaceuticals, Nutley, NJ) 0.5–1 mg PO every 8 hours as needed
Calcium channel blockers: Verapamil 80 mg three times daily	Triptans: Sumatriptan (Imitrex, GlaxoSmithKline, Brentford, Middlesex, UK) 25–100 mg PO at onset of symptoms, or 4–6 mg SC × 1, or 5 or 20 mg nasal spray × 1 Rizatriptan (Maxalt, Merck, Kenilworth, NJ) 5–10 mg PO × 1 Naratriptan (Amerge, GlaxoSmithKline, Brentford, Middlesex, UK) 1–2.5 mg PO × 1 Others
Tricyclic antidepressants: Amitriptyline 25–150 mg PO qhs, Nortriptyline (Pamelor, Mallinckrodt, Dublin, Ireland) 25–150 mg PO qhs, Protriptyline (Vivactil, Merck, Kenilworth, NJ) 5–10 mg PO three times daily	Combination: Single-tablet combination of sumatriptan and naproxen sodium (Treximet, Pernix Therapeutics, Morristown, NJ) 1 tab PO × 1

Abbreviations: PO, orally; qhs, before bedtime; SC, subcutaneously.

sumatriptan is helpful in some patients, but only for attacks lasting longer than one hour. Patients who develop chronic dysequilibrium may also benefit from vestibular rehabilitation therapy. Patients with perimenstrual migraine with vertigo often respond well to clonazepam (Klonopin; Roche Pharmaceuticals, Nutley, NJ) 0.5 to 1 mg PO every 8 hours as needed.

◆ Outcome and Follow-Up

Avoidance of identifiable triggers is highly successful. If pharmacotherapy is needed, the majority of patients respond favorably to prophylactic medications or triptans.

3.7 Tinnitus

◆ Key Features

- Tinnitus is the perception of sound without an unrelated external source of sound.
- Retrocochlear pathology in unilateral tinnitus must be excluded.
- Vascular anomalies and neoplasms should be excluded in pulsatile tinnitus.

Tinnitus is a common but poorly understood disorder. Its severity can range from insignificant to disabling. Tinnitus is a nonspecific symptom characterized by the sensation of buzzing, ringing, clicking, pulsations, and other noises in the ear. Objective tinnitus, or somatosounds, refers to noises generated from within the ear or adjacent structures. The term “subjective tinnitus” is used when the sound is audible only to the affected patient.

◆ Epidemiology

Fifty million Americans report tinnitus, representing an estimated prevalence of 10 to 15% of adults in the United States. Twelve million will seek medical attention. Males report it more frequently than females.

◆ Clinical

A targeted history at the initial evaluation of the patient, including risk factors for hearing loss, should be obtained to identify conditions that, if promptly identified and managed, can relieve tinnitus. It is important to determine whether the tinnitus is unilateral, pulsatile, new onset, or sudden onset with hearing loss. A history of hearing loss (unilateral, bilateral, asymmetric), noise exposure, ototoxic exposure, and head trauma should be elicited. Additional neurotologic symptoms such as vertigo, imbalance, dizziness, and even otalgia and otorrhea may help determine the etiology of the tinnitus. Depression, anxiety, and cognitive impairment often coexist with tinnitus. Tinnitus can also impair concentration, memory, and sleep. It is also important to elicit whether the tinnitus is mild, moderate, or severe based on effect on quality of life (i.e., how bothersome it is to the patient). Duration of tinnitus may facilitate discussion about natural history, treatment, and follow-up care. Persistent tinnitus occurs when symptoms last for ≥ 6 months.

Differential Diagnosis

Subjective (Nonpulsatile)

- Otologic: Hearing loss, otosclerosis, Ménière's disease
- Neurologic: Multiple sclerosis, trauma, vestibular schwannoma
- Infectious: Lyme disease, syphilis, otitis, meningitis

- Drug-induced: Aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, furosemide, vincristine, platinum-based chemotherapeutics
- Miscellaneous: TMJ or myofascial, dental, depression, idiopathic

Objective (Pulsatile)

- Turbulent flow: Glomus tumors, carotid atherosclerosis, benign intracranial hypertension, dural arteriovenous malformations, cerebral aneurysms
- Muscle contractions: Palatal myoclonus—stapedial, tensor tympani
- Spontaneous otoacoustic emissions

◆ Evaluation

Physical Exam

A complete otoneurologic exam should be performed. The mastoid, neck, and heart should be auscultated for vascular causes of tinnitus (e.g., heart murmurs, carotid bruits, vascular sounds). Focal neurologic signs, such as cranial nerve palsy, may suggest neoplastic cause, and additional evaluation and treatment are recommended. Otologic exam should include inspection of the external auditory canal, tympanic membrane, and middle ear for causes of conductive hearing loss that contributes to tinnitus. Otorrhea may be a sign of middle ear or external ear infection. The head and neck should be inspected and palpated for neoplasm. Occlusion of the ipsilateral internal jugular vein may alleviate pulsatile tinnitus in patients with venous sinus stenosis or diverticulum.

Imaging

Imaging of the head and neck should be considered in patients with unilateral tinnitus, pulsatile tinnitus, focal neurologic abnormalities, or asymmetric sensorineural hearing loss. Computed tomography (CT) or CT angiography (CTA) of the brain or temporal bone, or magnetic resonance imaging (MRI)/angiography of the brain or internal auditory canals are common choices for initial imaging studies for pulsatile tinnitus. When asymmetric sensorineural hearing loss or unilateral tinnitus is present, MRI of the internal auditory canal with and without contrast is considered to evaluate retrocochlear pathology as well as to rule out intracranial pathology.

Labs

Laboratory tests are not routinely ordered.

Other Tests

Comprehensive audiologic examination in patients is recommended. Auditory brainstem response (ABR) testing may be used to assess retrocochlear pathology. Tinnitus questionnaires (e.g., tinnitus handicap inventory, tinnitus functional index, tinnitus reaction questionnaire) may provide insight on how tinnitus affects quality-of-life measures.

Pathology

The pathophysiology of tinnitus is controversial. It is likely due to compensatory adaptation of the central auditory system to hearing loss. Although there is a clear association of tinnitus with hearing loss, many people with hearing loss do report tinnitus. Alterations in neurotransmitters along the auditory pathway have been linked to tinnitus.

◆ Treatment Options

The definition of tinnitus, its relationship to hearing loss, its association with ototoxic drugs, and the current pathophysiology of the disease should be explained to the patient. The management strategies should be discussed with patients with persistent, bothersome tinnitus. Additional approaches that can inform and educate patients include providing brochures, suggesting self-help books, describing counseling and sound therapy options, and discussing availability and limitations of pharmacologic and complementary and alternative medicines. Referral to support organizations (such as the American Tinnitus Association) and tinnitus specialists may also be helpful. It is important for patients to understand that there is no cure for tinnitus, but there are several treatments that can relieve the effects of tinnitus on concentration and sleep and improve hearing.

Medical

Hearing aid evaluation is recommended for patients with hearing loss and persistent, bothersome tinnitus. Sound therapy to mask tinnitus should be offered to patients with tinnitus that is bothersome and prolonged. Sound therapy device options include environmental enrichment devices (e.g., fans, TV, radio, tabletop sound generators) and hearing aids fitted with tinnitus maskers. The goal of sound therapy is tinnitus habituation, which is the adaptation of the auditory system that results in reduced intensity of perceived tinnitus. Tinnitus retraining therapy utilizes sound therapy for habituation but also incorporates simultaneous counseling. Identifying the frequency and intensity of the tinnitus may help tinnitus specialists individualize sound therapy for patients with more severe forms of tinnitus. Reassurance and noise precautions should also be provided.

Cognitive/behavioral therapy (CBT) has demonstrated effectiveness in treatment of tinnitus-related distress. It is usually performed by a mental health specialist. CBT teaches patients ways to identify negative thought patterns related to tinnitus and restructure to emphasize positive thought patterns. Treatment also includes behavioral interventions such as relaxation techniques, sleep hygiene, and auditory enrichment.

Antidepressants, anticonvulsants, anxiolytics, intratympanic medications, or dietary supplements (e.g., melatonin, zinc, *Ginkgo biloba*) should not be routinely recommended for persistent and bothersome tinnitus. Acupuncture may be considered as an alternative therapy in recalcitrant, bothersome tinnitus; however, its effectiveness has not been proven in clinical trial. Currently, there is no conclusive evidence that transcranial magnetic stimulation improves tinnitus, and therefore, it should not be recommended at this time.

Surgical

Where appropriate, surgery can be considered for some objective sources of tinnitus. Sigmoid sinus diverticula can be cauterized and dehiscences resurfaced via transmastoid approach. Vascular lesions (e.g., glomus tympanicum, glomus jugulare) causing tinnitus can be addressed with surgery and, in some cases, radiation or stereotactic radiosurgery. Tegmen dehiscences, meningoceles, and encephaloceles can be repaired through a middle cranial fossa and/or transmastoid approach. Arteriovenous malformations can be embolized or surgically resected.

◆ Outcome and Follow-Up

Healthcare professionals have the opportunity to help tinnitus patients learn how to cope with and manage their tinnitus. Realistic expectations and treatments should be provided to the patient. Annual audiograms should be performed to determine progression of hearing loss.

3.8 Cerebellopontine Angle Tumors

◆ Key Features

- The most common tumors of the cerebellopontine angle are vestibular schwannomas, followed by meningiomas and epidermoids.
- Treatment options depend on the etiology but may include observation, stereotactic radiosurgery, and surgical excision.
- Surgical approaches to the cerebellopontine angle are the middle fossa, retrosigmoid, and translabyrinthine approaches.

The cerebellopontine angle (CPA) is a cerebrospinal fluid (CSF)-filled space found at the ventral aspect of the junction between the cerebellum and the pons. Most neoplasms found in this location are benign. Vestibular schwannomas (also known as acoustic neuromas) are the most common, followed by meningioma, and epidermoids. Presenting symptoms usually include unilateral sensorineural hearing loss, tinnitus, and dizziness/imbalance.

◆ Epidemiology

Vestibular schwannomas are the most common CPA neoplasm. These most commonly arise from the inferior vestibular division of CN VIII. Spontaneous yearly occurrence is ~1 in 100,000, or roughly 2,280 new cases annually in the United States. Most tumors grow slowly (average is 1–2 mm per year). Bilateral tumors can be seen in patients with neurofibromatosis 2 (NF2).

◆ Clinical

Signs and Symptoms

Presenting symptoms can include unilateral hearing loss, unilateral tinnitus, or progressive imbalance or vertigo. Difficulty talking on the phone with one ear is a common complaint. Sudden hearing loss may be the presentation in ~10% of cases. Large tumors can cause brainstem compression and cranial nerve (CN) V, VI, VII, VIII, IX, X, and XI palsies. Compression of the fourth ventricle may cause hydrocephalus, typically with giant tumors measuring more than 4 cm. Ataxia, altered mental status, headache, nausea, vomiting, diplopia, respiratory depression, and coma can also occur.

Differential Diagnosis

The differential diagnosis may include vestibular schwannoma, meningioma, epidermoid, arachnoid cysts, facial nerve schwannoma, trigeminal schwannomas, endolymphatic sac tumors, chondrosarcoma, and metastatic tumor.

◆ Evaluation

Physical Exam

A complete otoneurologic, head and neck, and full cranial nerve exam should be performed. Cranial nerve deficits should be documented. The corneal reflex (CN V₁) and blink reflex (CN VII) can be assessed using a wisp of cotton. Hoarseness and dysphagia warrant fiberoptic laryngoscopy for evaluation. Spontaneous, gaze, and headshake nystagmus tests may indicate unilateral vestibular weakness (fast phase to the opposite side). Fall or sway may be seen with Romberg, Fukuda stepping, and tandem gait tests. Dysmetria may be seen on cerebellar tests such as the finger-to-nose test. Hitzelberger sign may also be elicited (reduced sensation of the posterior ear canal from tumor compression of the facial nerve).

Imaging

Magnetic resonance imaging (MRI) of the brain with and without contrast with thin cuts through the internal auditory canals is the gold standard for diagnosis. Vestibular schwannomas and meningiomas are slightly hypointense to isointense on T1-weighted images, are heterogeneously hyperintense with cystic areas on T2, and enhance with contrast. Meningiomas are more sessile, broad-based, and may display a dural tail. They are rarely centered within the internal auditory canal. Diffuse and extensive dural involvement can occur ("en plaque meningioma"). Computed tomography (CT) scan of a meningioma may demonstrate calcifications and adjacent hyperostosis. Epidermoids are usually hypointense on T1 and hyperintense on T2. They do not enhance with contrast but they may be hyperintense on fluid-attenuated inversion recovery (FLAIR).

Other Tests

- Audiogram: Unilateral pure-tone sensorineural loss is usually seen. Decreases in speech discrimination are common and usually greater than

Table 3.15 Management of non-neurofibromatosis Type 2 vestibular schwannomas (acoustic neuromas)

Tumor size	Options
Small (≤ 2 cm diameter), noncystic	Observation ○ MRI every 6–12 mo until stable, then annually ○ Annual audiometry ○ Treat if > 2 mm interval growth RadiotherapySurgery ○ Translabyrinthine approach if no useful hearing ○ Suboccipital or middle fossa approach if good discrimination score
Large (> 2 cm diameter)	Treat (consider age comorbidities to determine best modality)

expected considering pure tones. Decay in speech discrimination may be demonstrated at higher test stimuli (rollover).

- Auditory brainstem response (ABR) test: The sensitivity of ABR is approximately 95%; however, this percentage drops to 67% for smaller, intracanalicular tumors. Interaural difference in wave V of > 0.2 ms is abnormal.
- Vestibular testing: Tumors may cause weakness of the superior and/or inferior vestibular nerves. Peripheral vestibular weakness may be demonstrated on caloric testing (function of horizontal canal and its afferent pathway through the superior vestibular nerve) and vestibular evoked myogenic potentials (VEMP). Cervical VEMP may detect dysfunction of the saccule or inferior vestibular nerve; ocular VEMP may detect dysfunction of the utricle or superior vestibular nerve.

Pathology

On pathology, vestibular schwannomas demonstrate Antoni A (hypercellular) and Antoni B (hypocellular) areas. Verocay bodies are fusiform cells arranged in palisades, characteristic of Antoni A areas. Meningiomas demonstrate round uniform nuclei with intranuclear pseudoinclusions and eosinophilic cytoplasm. Psammoma bodies may also be present. Epidermoid cysts demonstrate squamous epithelium.

◆ Treatment Options

See **Table 3.15** for treatment options for a vestibular schwannoma. Tumor size, growth rate, severity of symptoms, hearing levels, anesthetic risk, patient age, and other factors must be evaluated for the individual patient to determine the best option for treatment. The average growth rate of a vestibular schwannoma is ~1 to 2 mm per year, although this can vary. Treatment options include observation with serial MRI scans every 6 to 12

months (to determine growth rate), radiotherapy (e.g., stereotactic radiosurgery), and microsurgical resection.

Stereotactic radiosurgery, as by Gamma Knife (Elekta, Stockholm, Sweden) or CyberKnife (Accuray, Inc., Sunnyvale, CA), can deliver a high dose of radiation to a small area of the body with excellent precision. Radiation injury to surrounding normal tissue is reduced compared to conventional fractionated radiation. Facial nerve palsy occurs in ~3% of patients. The hearing preservation rate is ~23% at 10 years. MRI scans are obtained at 6- or 12-month intervals to assess for tumor growth. Stereotactic radiosurgery is generally reserved for tumors that are <2.5 cm, although indications may still be expanding.

Microsurgical approaches for resection of vestibular schwannoma are the middle cranial fossa, retrosigmoid, and translabyrinthine approaches (**Fig. 3.13**). The preference and experience of the surgeon will factor into the decision regarding the approach. Meningiomas have a slightly better chance of hearing preservation than vestibular schwannomas.

- The middle fossa approach is appropriate for small tumors less than 1.5 cm (mostly intracanalicular). Hearing preservation is possible, and the entire internal auditory canal (IAC) can be accessed. Air cells need to be obliterated with bone wax.

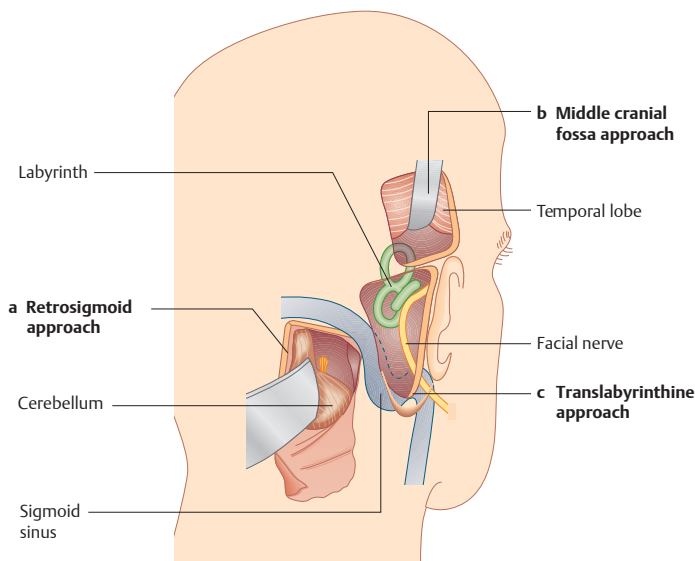


Fig. 3.13 Surgical approaches to the lateral skull base. The retrosigmoid approach requires a craniotomy posterior to the sigmoid sinus. The middle cranial fossa approach requires a transtemporal craniotomy and retraction of the temporal lobe. The translabyrinthine approach is a transmastoid approach that requires mastoidectomy and labyrinthectomy to expose the internal auditory canal and posterior cranial fossa. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:301.)

- The retrosigmoid (suboccipital) approach can be performed for small and large tumors. Hearing preservation is possible; however, only the medial two-thirds of the IAC can be exposed safely. Cerebellar retraction may cause cerebellar edema and brainstem compression; however, decompression of CSF at the foramen magnum can reduce the need for cerebellar retraction. Air cells need to be obliterated with bone wax.
- The translabyrinthine approach can be used for any size tumor, but hearing preservation is not possible. Decompression of the labyrinthine segment of the facial nerve improves visualization and dissection of the meatal segment of the facial nerve. Cerebellar retraction is usually not necessary. The auditory (eustachian) tube is obliterated with fascia or bone wax. The surgical defect is generally packed with abdominal fat.

The management of vestibular schwannoma is continually evolving. Large vestibular schwannomas compressing the brainstem require surgical resection; patients who cannot tolerate general anesthesia may receive radiosurgery as the only alternative. Observation is offered for small to medium-sized vestibular schwannoma in all age groups. Microsurgical resection is generally preferred for growing small to medium-sized vestibular schwannomas in young, healthy patients (because of long-term complications of radiation exposure); radiosurgery may be a better alternative in older patients with medical comorbidities. Hearing preservation surgery can be performed through the middle cranial fossa or retrosigmoid approaches depending on the size and location of the tumor; however, the middle cranial fossa should be avoided in patients > 65 years due to higher complication rate from temporal lobe retraction. Translabyrinthine and retrosigmoid approaches can be performed in patients with small to large tumors; there is a potential for hearing preservation using the retrosigmoid approach with small to medium-sized tumors.

NF2 patients with bilateral vestibular schwannomas require special consideration and sometimes aggressive management to avoid eventual bilateral deafness. Enlarging tumors may impair function of the facial and cochlear nerves bilaterally. Unlike most sporadic acoustic tumors, NF2 tumors often infiltrate nerve fibers rather than displace them. Debulking procedures may not be feasible. Managing multiple cranial neuropathies can be challenging; lower cranial nerve involvement is common. Meningiomas have a slightly better chance of hearing preservation. NF2 patients require individualized evaluation and treatment planning. Hearing can potentially be aided with an auditory brainstem implant during vestibular schwannoma surgery.

◆ Complications

See **Table 3.16** for timetable and description of possible complications, as well as management issues. Initial postoperative facial weakness should be treated with steroids and eye protection in the short term. Appropriate eye

Table 3.16 Timetable of possible postoperative complications following craniotomy and cerebellopontine angle tumor resection

← ABCs →						
Airway compromise: Mucus plugging, ventilator problems, vocal fold paralysis from lower cranial nerve dysfunction Bleeding: Within the CPA from operative field—contralateral long-tract signs, ↑ ICP Circulation: Normotension not maintained						
← Intracranial hemorrhage → (Parenchymal, epidural, subdural) ← Infarct → (AICA, cerebellar, cerebral) CNS edema ← SIADH/DI → (Watch sodium; consider hydrocephalus; RAS suppression; medication effects) ← Emesis followed by change in mental status → (Bleed, pneumocephalus, CSF leak. Could be prompted by movement, or disorientation in CT scanner) ← Fever, meningitis → ← DVT, PE → (TEDS, sequential compression stockings)						
TIME: →						
Postoperative day #						
0	1	2	3	4	5	+
	ICU	floor	Start PO intake Start PT			
Comments: <i>RAS suppression:</i> Edema, increased pressure or displacement can inhibit brainstem reticular activating system function, causing, decreased consciousness in the early postoperative period. CPA surgery and cerebellar retraction predispose the patient to edema, especially 24–72 hours postoperatively. Keep Na > 140; consider steroids. <i>Increased intracranial pressure:</i> Hemiparalysis with obtundation, a fixed dilated pupil and respiratory distress are the hallmarks of rapidly increasing ICP with peduncular herniation. If the patient does not recover properly from anesthesia or there is any unexpected neurologic abnormality, suspect hematoma, hydrocephalus with increased ICP, or cerebellar infarct. If deterioration is slow, obtain a CT scan; if rapid, go to OR. <i>Meningitis:</i> Most common at postoperative day 4 or later; marked by postoperative fever, increased headache, malaise, photophobia, nausea. Promptly obtain LP with Gram stain, cell count, glucose, cultures. Treat with IV antibiotics and close monitoring. <i>CSF leak:</i> There is a 5–17% incidence. Mastoid air cells may be opened intraoperatively during craniotomy or IAC drilling. If CSF rhinorrhea develops, treat with antibiotics, head of bed elevation, ± intermittent LP versus lumbar drain. If copious leakage, prompt mastoidectomy with fat obliteration is indicated.						

Abbreviations: AICA, anterior inferior cerebellar artery; CNS, central nervous system; CPA, cerebellopontine angle; CSF, cerebrospinal fluid; CT, computed tomography; DI, diabetes insipidus; DVT, deep vein thrombosis; ICP, intracranial pressure; ICU, intensive care unit; LP, lumbar puncture; OR, operating room; PE, pulmonary embolus; PT, physical therapy; RAS, reticular activating system; SIADH, syndrome of inappropriate antidiuretic hormone; TEDS, thromboembolic deterrent stockings.

care is critical to avoid exposure keratitis, which is entirely preventable. A recommended regimen includes hourly artificial tears drops while awake and Lacri-Lube ointment when sleeping (Allergan, Inc., Irvine, CA). Some advocate use of a small square of Saran Wrap (SC Johnson, Racine, WI) taped over the eye while sleeping ("moisture chamber"). The ointment and wrap can also be used while awake if the patient is noncompliant with the drops regimen.

Moreover, in the setting of concurrent facial and trigeminal dysfunction, the patient will not feel the pain of the eye drying or abrading. Prolonged facial weakness may warrant an upper eyelid gold weight placement, as well as surgical reanimation. If no function returns after 18 to 24 months, then static or dynamic reanimation is recommended (see also **Chapter 9.2**). Gracilis free flap can be offered for facial reanimation in the appropriate patient.

CSF otorrhea, CSF rhinorrhea, or incisional CSF leaks may resolve with bed rest, stool softeners, lumbar drain placement, and pressure dressings. Persistent CSF leaks should be treated with reexploration, middle ear obliteration, and auditory tube closure/obliteration.

Unilateral complete hearing loss can be rehabilitated with a contralateral routing of signals (CROS) hearing aid or Bi-CROS hearing aids. Osseointegrated bone conducting hearing aids for single-sided deafness can improve speech understanding in noise (see **Chapter 3.5.5**). These techniques function on the principle of transferring sound from the deaf ear to the hearing ear.

◆ Outcome and Follow-Up

Patients may have significant vertigo following surgical treatment until adequate compensation occurs over several weeks. This is especially true of patients without preoperative vestibular hypofunction, who therefore have no previous compensation. Short-term vestibular suppressants and vestibular therapy can help improve symptoms. Careful monitoring of facial nerve function and observation for evidence of meningitis or CSF leak is critical. CSF leak rates run from 10% to 15%. Facial nerve preservation rates are dependent on both surgeon experience and tumor size. Facial nerve preservation can be more difficult for previously irradiated tumors.

Patients undergoing observation or radiation should have serial MRI scans every 6 to 12 months to watch for tumor growth. Surgical patients should have follow-up MRIs annually, which may become more infrequent with time if no growth or recurrence is identified.

3.9 Superior Semicircular Canal Dehiscence Syndrome

◆ Key Features

- Superior semicircular canal dehiscence (SSCD) occurs when there is an absence of intact bone overlying the superior semicircular canal.
- Presentation can include the Tullio phenomenon, autophony, and hyperacusis.
- A third mobile window leads to low-frequency conductive loss with intact acoustic reflex.
- Vestibular-evoked potentials (VEMP) can demonstrate reduced thresholds and large peak-to-peak amplitudes.
- Treatment is primarily surgical (resurfacing and/or canal plugging).

Superior semicircular canal dehiscence (SSCD), or semicircular canal dehiscence syndrome (SCDS), was first described by Minor in 1998 and is defined by the absence of bone overlying the superior semicircular canal, leading to vestibular and auditory symptoms. Patients with SSCD can present with conductive hearing loss (CHL) with intact acoustic reflexes, conductive hyperacusis, pulsatile tinnitus, and Tullio phenomenon.

◆ Epidemiology

The incidence of SCDS is unknown. However, temporal bone studies suggest that the anatomic incidence of dehiscence of the superior canal is 0.5%.

◆ Clinical

Signs and Symptoms

Clinically, signs and symptoms may range from very mild to severe. Patients with SSCD may exhibit some or all of the following:

- Tullio phenomenon: Vertigo or dizziness induced by loud sound, with oscillopsia and nystagmus
- Autophony: The person's own speech or other self-generated noises, such as chewing, are perceived as unusually loud in the affected ear.
- Dizziness, disequilibrium, or true vertigo
- Hyperacusis
- Conductive hearing loss, often in the low-frequency range
- Aural fullness
- Pulsatile tinnitus
- Headache

Differential Diagnosis

With a healthy or normal-appearing middle ear, the presence of normal acoustic reflex should help discriminate otosclerosis with stapes immobility from possible SSCD. Obviously, one must exclude other identifiable causes of CHL, such as tympanic membrane perforation, ossicular chain problems, or middle ear effusion. The differential diagnosis for dysequilibrium is broad. Noise-induced vertigo raises suspicion for SSCD. Perilymphatic fistula and an enlarged vestibular aqueduct are possible conditions associated with an inner ear third mobile window. Ménière's disease, patulous auditory (eustachian) tube, and migraine may have some overlapping findings.

◆ Evaluation

History

History should focus on complaints just described, including history of dizziness in loud noise, bothersome autophony, or aural fullness.

Physical Exam

An otoneurologic exam is performed to rule out other etiologies of conductive hearing loss and dizziness. Tullio phenomenon and Hennebert sign may be elicited. Tuning fork tests may indicate a conductive hearing loss. Conductive hyperacusis may be perceived when a tuning fork is placed on the medial malleolus of the ankle. Nystagmus may be elicited with Valsalva against a closed glottis or pinched nostrils or with tragal compression.

Imaging

High-resolution thin-cut computed tomography (CT) scanning (0.6-mm slice thickness or less) of the temporal bone is obtained and can suggest dehiscence of bone overlying the superior semicircular canal by examination of the coronal images. Dehiscence may be better appreciated if the images can be reformatted in the plane parallel to the superior canal (Pöschl) and in the plane perpendicular to the superior canal (Stenvers view).

Labs

Laboratory testing is not routinely ordered for SSCD.

Other Tests

Audiologic Assessment

Audiologic assessment should include pure tone testing for air conduction thresholds from 250 to 8,000 Hz, and bone conduction thresholds from 250 to 4,000 Hz. Conductive hyperacusis may be assessed by administering bone conduction testing at stimulus levels less than 0-dB hearing loss (e.g., -5 to -10 dB). Assessment should also include tympanograms, otoacoustic emissions, and acoustic reflexes.

Vestibular Evoked Myogenic Potential Testing

VEMP testing can be used to provide evidence of altered vestibular function; thus, it can help to determine if a suspected dehiscence of the superior canal, seen on CT scanning, is associated with an alteration of inner ear physiology. Patients with SSCD have larger peak-to-peak amplitudes and lower thresholds on VEMP testing. In cervical VEMP, sound is delivered via insert earphones, and activity is recorded from the ipsilateral sternocleidomastoid muscle. In ocular VEMP, activity is recorded from the contralateral inferior oblique muscle. Some studies suggest that ocular VEMP is more sensitive for detecting SSCD than cervical VEMP.

Pathology

A thinning of the bone overlying the superior semicircular canal leads to a third mobile window within the inner ear. This alters inner ear mechanics, leading to the clinical findings. The precise cause of the dehiscence is unknown.

◆ Treatment Options**Medical**

Accurate diagnosis is important, as unnecessary stapes surgery (for the CHL) must be avoided. If the patient has mild symptoms, the identification of a source for the symptoms may provide reassurance, and ongoing observation may be all that is required.

Surgical

If the patient has severe symptoms, surgery to resurface or occlude the dehiscence may be warranted. The surgical candidate should have a correlation of symptoms, exam findings, VEMP, and imaging pointing to SSCD. The procedure can be performed via middle cranial fossa or transmastoid approaches. The middle cranial fossa allows for excellent exposure of the dehiscence; resurfacing and/or plugging can be performed under direct visualization. Hydroxyapatite cement and bone pate is subsequently placed. However, in the middle cranial fossa approach, the temporal lobe is retracted superiorly, which may cause temporal lobe edema and seizures. Dural tear, cerebrospinal fluid (CSF) leak, and injuries to the superficial middle cerebral vein (vein of Labbé) are additional risks. Facial nerve monitoring is performed, and intravenous steroids and mannitol are administered. Patients are admitted to the neurosurgical ward for 2 or 3 days. Temporomandibular joint (TMJ) pain is common.

The dehiscence is not directly visualized with transmastoid approaches. A low-lying dura may preclude repair through this approach. The otic bone may need to be drilled to gain access to the superior semicircular canal; crura on either side of the dehiscence are plugged with bone wax, fascia, and/or bone dust/fragments. The superior aspect of the superior semicircular canal can be resurfaced using cartilage or cortical bone inserted through a bony window in the tegmen mastoideum; a tear in the membranous labyrinth of the superior semicircular canal can occur unnoticed due to limited visualization. Patients may be admitted for symptomatic control of dizziness and/or nausea.

◆ **Complications**

Global vestibular hypofunction and sensorineural hearing loss from perilymphatic fluid leak or labyrinthitis after the surgical intervention can be treated with systemic and/or intratympanic steroids. Facial weakness from inadvertent avulsion/injury to the greater superficial petrosal nerve in the middle cranial fossa approach should be treated with high-dose oral steroids. CSF leaks should be treated with bed rest, stool softeners, and lumbar drain; surgical intervention may be necessary if CSF leak persists.

◆ **Outcome and Follow-Up**

Most patients demonstrate improvement in autophony, conductive hearing loss, and dizziness. Vestibular therapy can be helpful postoperatively. Postoperative hearing testing is obtained several weeks after surgery. If symptoms return, reimaging and repeat VEMP testing should be considered.

[REDACTED]

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